

INSIGHTWELLNESS-AI

Milestone 2

Pivot from milestone 1

- **Feedback from MS1:** Model is a sophisticated BMI rule
- **Pivot:** Removed Height & Weight features; focus on lifestyle features. The goal of our approach is to detect persons at risk before the weight gain.

Model Comparison (1/2)

Model	Accuracy	F1-macro	CV F1 mean	CV F1 std
Current Model	83.92%	83.98%	78.89%	±4.9%
Height/Weight Model	96.45%	96.33%	96.18%	±0.93%

Model comparison (2/2)

Top 5 feature importances

Height/Weight Model

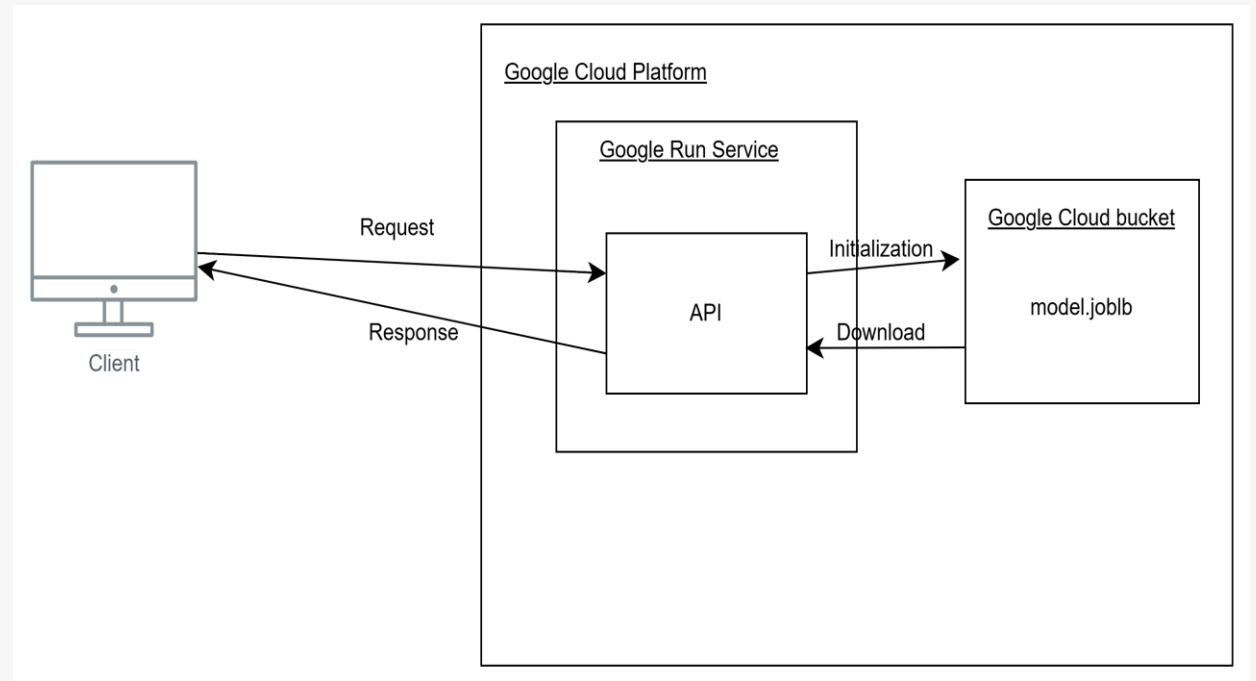
Feature	Importance
Weight	51.94%
Height	17.04%
FCVC	9.09%
Gender	7.20%
Age	3.81%

Current Model

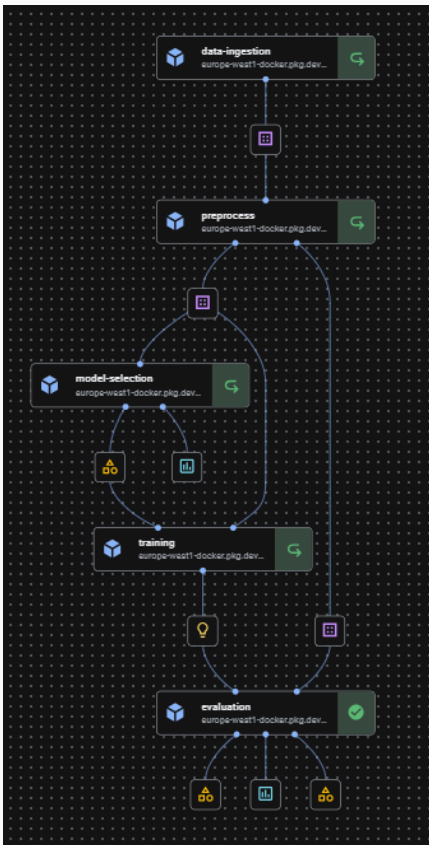
Feature	Importance
Age	17.71%
FCVC	15.32%
TUE	13.11%
NCP (# meals)	8.74%
FAF	8.53%

API Architecture

- Flask API with 4 endpoints: /, /status, /features, /predict
- Input validation (feature schema, ...)
- **Base Image:** Built using python:3.11-slim to minimize container size
- **Dependency Management:** Managed via **UV**, ensuring lightning-fast, reproducible builds.
- Runs locally and in cloud
- **Cost Management:** Scales to zero (Max. 50)
- **Publicly accessible**



Vertex AI Pipeline



- **Data ingestion:** Collect data from BigQuery
- **Preprocess:** Cleaning, feature engineering and train/test split.
- **Model selection:** Hyperparameter tuning.
- **Training:** Train on train data.
- **Evaluation:** Produces performance metrics using test data.

Next steps

- Build Streamlit dashboard and connect it to the API.
- Complete SHAP explanation endpoint
- Update GitHub CI/CD pipeline to automatically build, push and run the Vertex AI pipeline and the API.
- The model is in a GCS bucket, use Vertex AI model registry instead.
- Explore Terraform.
- Explore Locust.
- Work on unit tests.